micro:bit

Introduction to Computer Science

Mini project

Microsoft

Lesson A
Looking back so far

What we'll learn in Lesson A

- A review of the computer science concepts we have covered so far
- A chance to brainstorm

Microsoft

Project overview

You will use the design thinking approach to create an original project that serves a purpose by:

- Solving a problem
- Filling a need

Show what you know

- Learn and demonstrate something new
- Include a maker component

Concepts to include

Demonstrate the use of the following concepts:

- Design thinking
- Input/processing/output
- Variables
- Conditional statements
- Iteration/loops

Concepts review

1. Making
2. Processing and algorithms
3. Variables
4. Conditionals
5. Iteration and looping

Name one thing you've learned from each of these lessons.

Making

- The micro:bit is very effective at bringing real things to life.
- It can be supported in a cardboard holder, attached to a wand, or even sewn into fabric.
- The design thinking process is a helpful way to create an optimal solution that will solve a problem or address a need.

The design thinking process

Step 1

Empathize

Learn about your audience

Step 2

Define

Construct point of view based on user needs

Step 3

Ideate

Brainstorm and come up with creative solutions

Step 4

Prototype

Build representation of your ideas

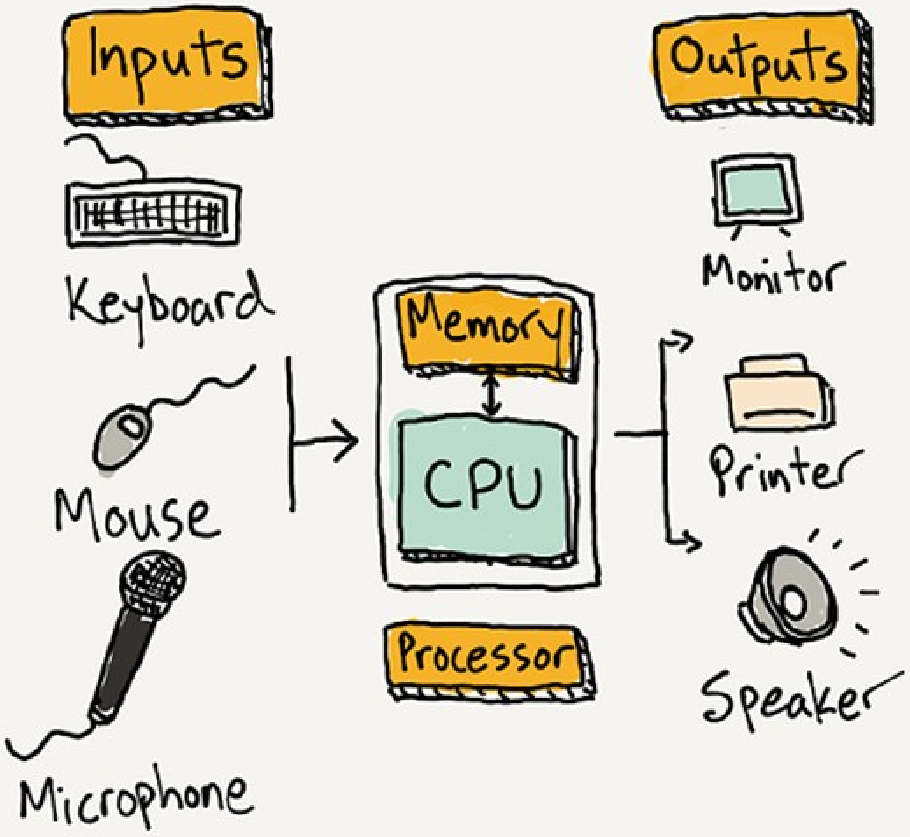
Step 5

Test

Test your ideas

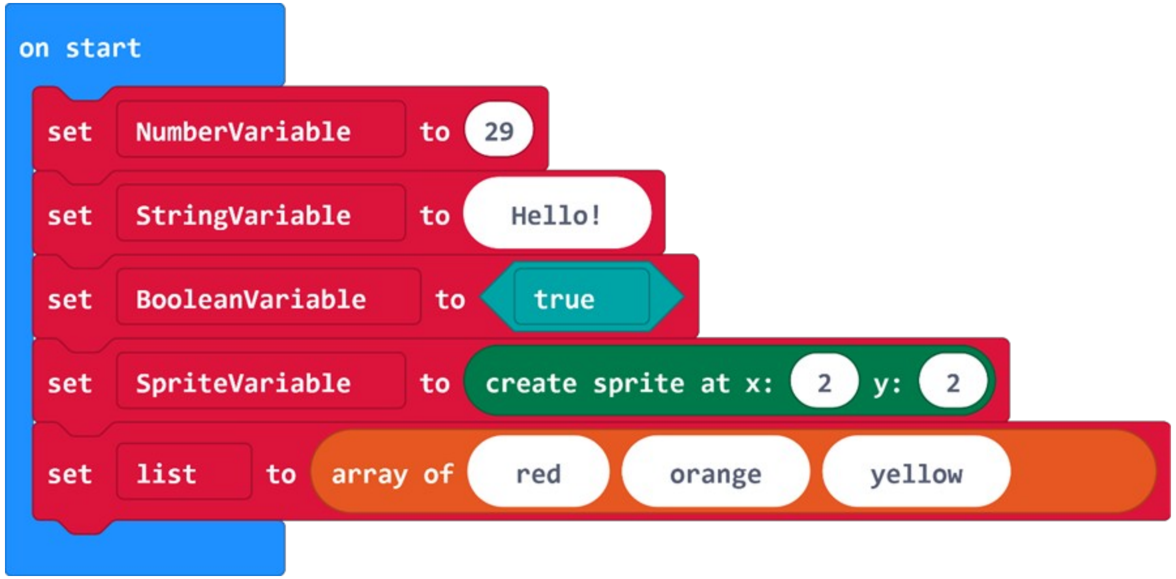
Processing and algorithms

- There are 4 parts to a computer:
  - Processor
  - Memory
  - Inputs
  - Outputs
- The code you write for the micro:bit processes data from its inputs, then outputs it in some way.
- An algorithm is a series of specific instructions, or steps, that solves a problem or accomplishes a task.



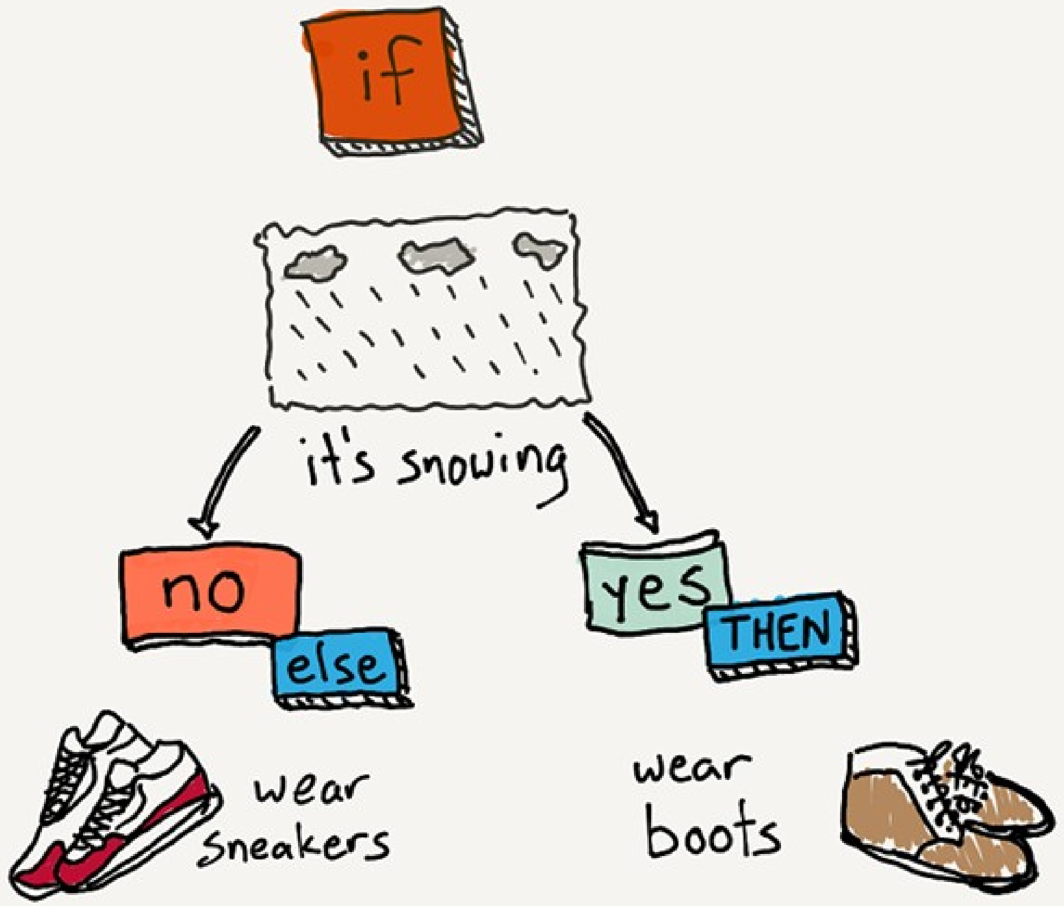
Variables

- Variables store information so that it can be accessed or referenced later.
- Some variables hold information that changes, and some hold information that stays constant.
- It is important to name your variables with something that explains what type of information it holds.
- Variables can hold information of different types: Numbers, Strings (text), Boolean (true/false), Sprites, or Arrays.



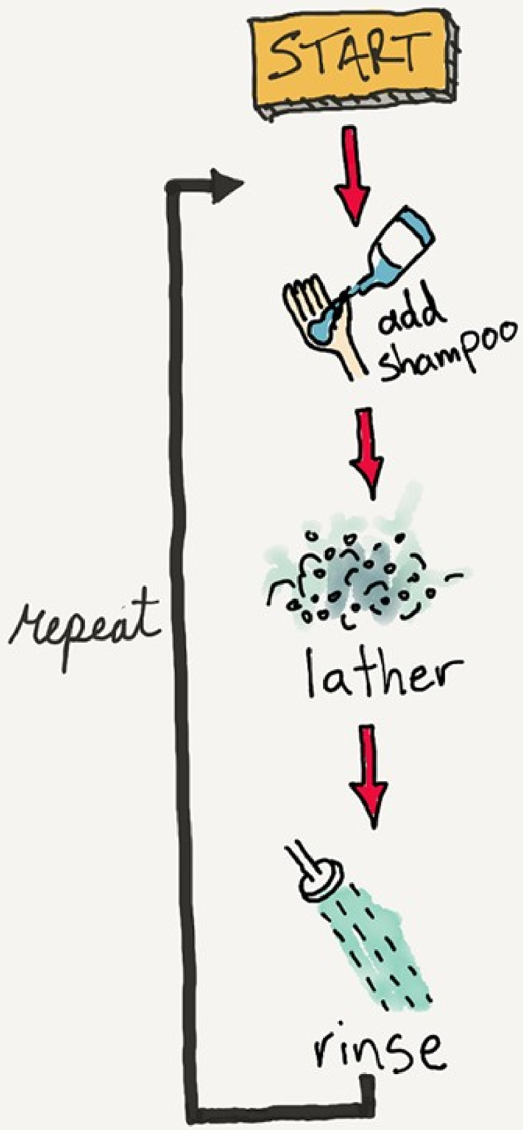
Conditionals

- A conditional statement is a rule or criteria that must be met for an action to be carried out.
- Conditional Statements are written in the form of **IF… THEN…** and sometimes **IF… THEN… ELSE…**
- Conditionals are used to tell a computer when to do something.



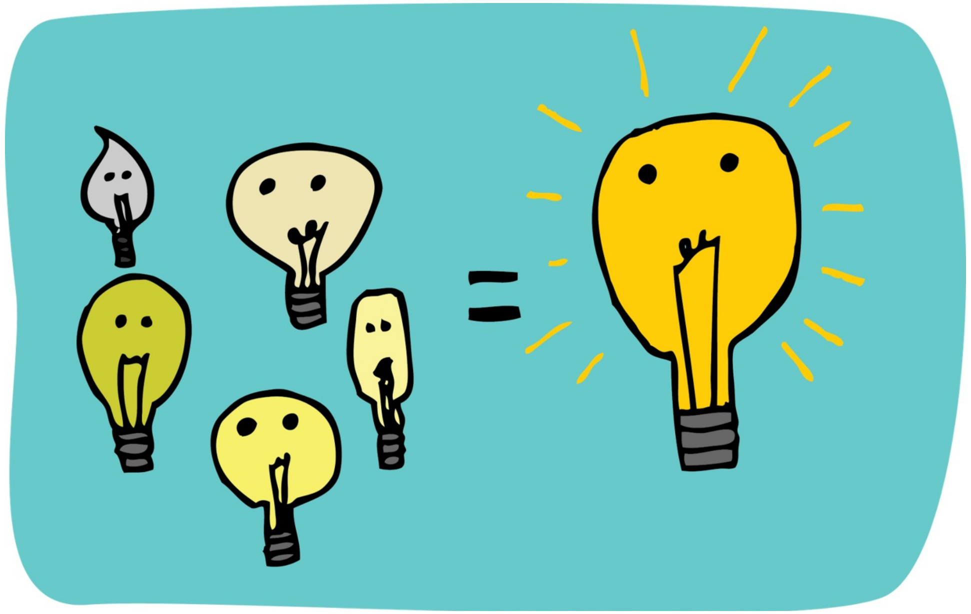
Iteration and looping

- **Iteration** means repeating a process or sequence of steps.
- A **loop** allows you to repeat lines of code, making your programs more efficient.
- There are **3 types of loops** in MakeCode:
    - **Repeat loop:** Repeat a certain number of times.
    - **For loop:** Repeat based on an index variable which increments each time.
    - **While loop:** Repeat until a certain condition is met.



Mini project brainstorm

- You will be creating a mini project with the micro:bit that **incorporates all the concepts** we've covered so far in class.
- Get in **groups of 4** to **brainstorm** ideas for your mini projects. Think about all the concepts we've covered so far and how your project could solve a problem or fill a need.



What we learned in Lesson A

- A fresh look at the computer science concepts we have covered so far
- Brainstormed for your mini-project!

Next time: Lesson B

- Code and make a mini project

Lesson B

Code and make a mini project

What we'll learn in Lesson B

- Design a project and build it

Mini project

Design a project by yourself that serves a purpose by:

- Solving a problem
- Filling a need
- Show what you know
- Learn something new
- Include a maker component

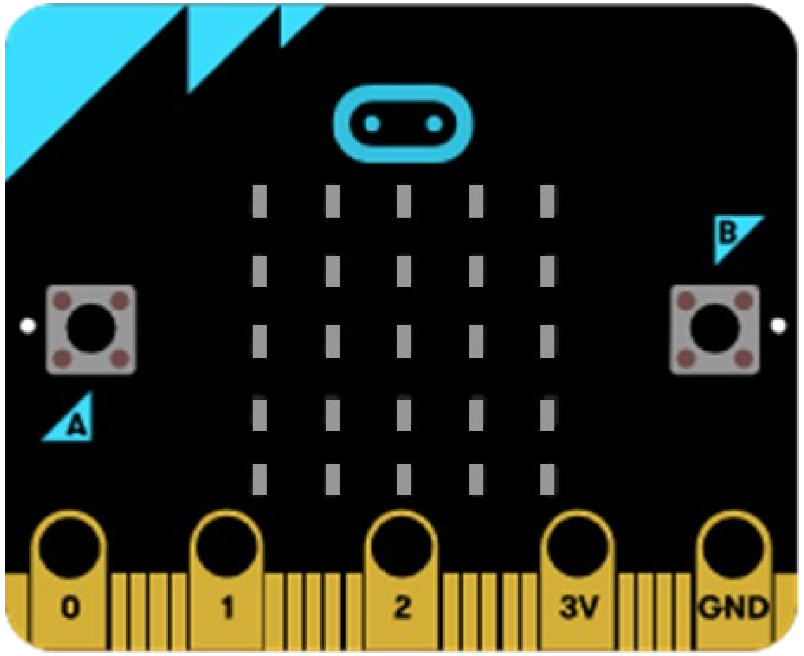
Project expectations

Use a design thinking approach and ensure the project meets these specifications:

- Create an **original project using the micro:bit**
- Incorporate a **physical component** to the project.
- Demonstrate the use all of the following concepts:
  - **Design thinking**
  - **Input/Processing/Output**
  - **Variables**
  - **Conditional statements**
  - **Iteration/loops**

Project ideas

- New and improved fidget cube
- Moving monster
- Musical instrument
- Fishing game
- Air guitar
- Screensaver
- Interactive book
- Game
- Binary clock or some other way to represent numbers visually

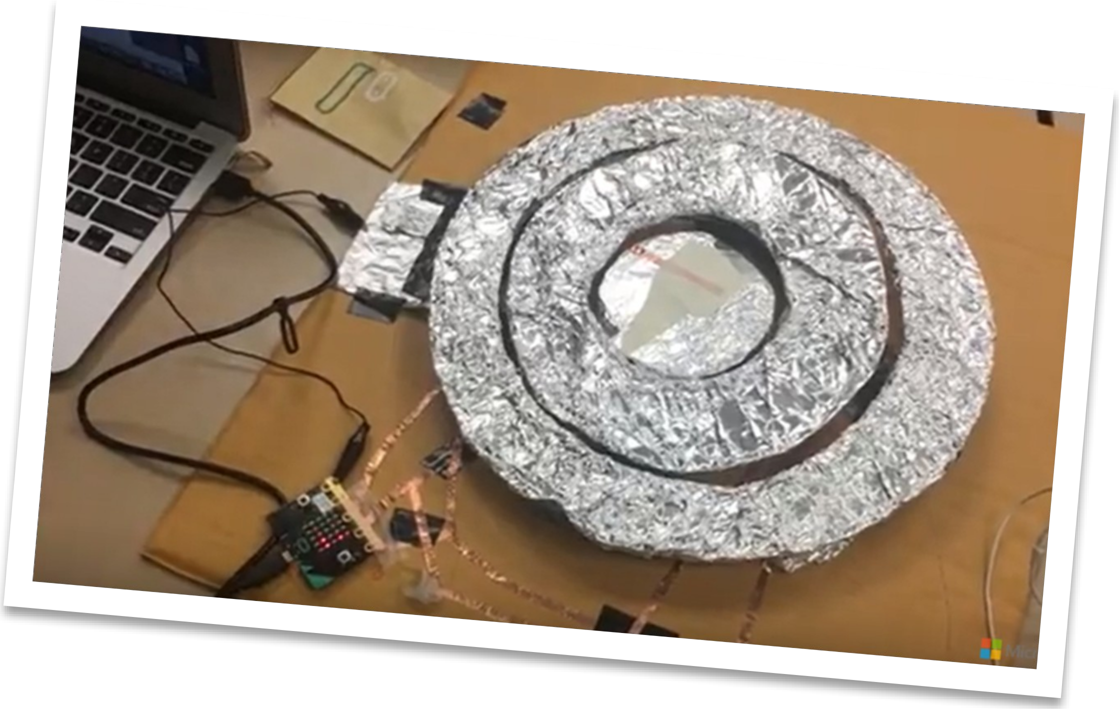


Project example:
Toss the ball

- This is a skill game in which an aluminum foil ball is thrown into a plastic cup.
- Copper tape lining the sides and bottom of the cup completes the circuit when the ball touches it.

Project example: micro:bit bullseye

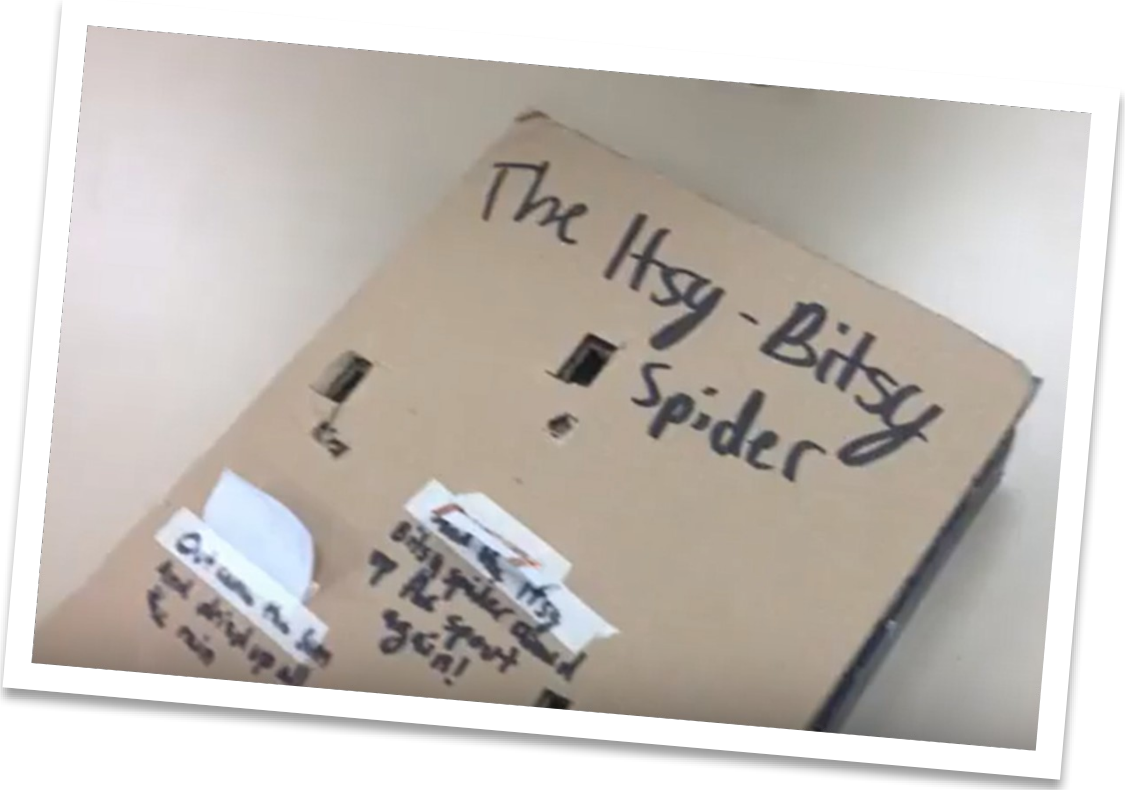
- This is a skill game in which tennis balls are thrown underhand at one of the three rings, which are lined with aluminum foil so they complete a circuit underneath when the ball makes contact with the ring.
- See it in action at youtu.be/NZUpoSixf4E



Project example: micro:bit storybook

This is a prototype of a storybook that could use the micro:bit to display animations for part of the story.

- Copper tape is used on the underside of the paper flaps to make contact between the GND pin and each of the other pins in sequence.
- See it in action at youtu.be/yg1NNLMqa9c



What we learned in Lesson B

- Designed a project and built it

Next time: Lesson C

- Mini project showcase!

Lesson C

Mini project showcase

What we'll learn in Lesson C

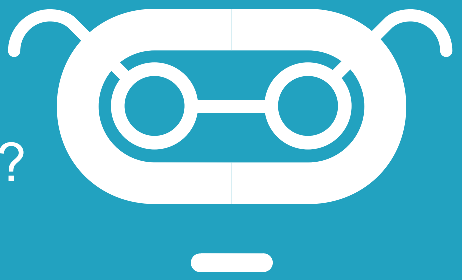
- The amazing inventions everyone made!

Showcase

- Give a demo of how it works
- Show how the code works
- Show how you assembled the physical/maker component
- Tell us how you used one concept you knew
- Tell us how you learned something new

Reflection Diary

- Talk about one challenge you faced in creating this project. How did you overcome this challenge?
- What did you demonstrate that you already knew?
- What was the new thing you learned in order to make this? How did you learn about it?
- Who in the class provided help to you along the way? How?
- Describe one specific thing you are proud of in this project.
- What would you do differently next time?
- If you had another week to work on this project, what might you add or improve?
- Publish your MakeCode program and include the link.



Sample Reflection Diary (excerpt)

I spent this week finishing up little details with my program, making it work better and more user friendly. The part that surprised me the most was the little things that kept popping into my head, little suggestions that could potentially be good to add, but might not be necessary or even useful. At the beginning of the assignment, I just added them as quickly as I thought of them, but as the project neared the midpoint and conclusion, I found myself considering if I actually needed them (as previous additions have since been quickly deleted).

Another thing that I find interesting about this is that it is a rather specialized project. Not many people would have a reason to use it except for me. However, this is supposed to be easily used by other people, so I have to take them into consideration as I design the project. I also realized that I had, at some point, broken part of my code without noticing it, so I now have to fix part of it. The reason that it is a problem is because I added a lot of code at once without testing it, which is unfortunate. Next time, I will add small amounts of code and test it first.

What we learned in Lesson C

- That we know a lot!
- That we are all super creative!